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Genetic structure of wild populations of Drosophila melanogaster*:* Wild flies of *D. melanogaster* were trapped from different localities near Izatnagar and Bareilly. Of these two populations were established. To study the genetic equilibrium conditions of these two populations, a nested design was set up.

Thirty sires selected at random were each mated to five dams; from each dam ten progeny were to be scored for egg production, per cent hatchability, total number of adults and sex ratio.

The model used was as follows :

$X_{ijk}=\mu+S_i+d_{j+(i)}+f_{k(ij)}$ where X_{ijk} is the measured performance of kth progeny of jth dam mated to ith sire.

S_i, effect of ith sire on its progeny presumed to be mostly genetic NID (O, O^2_s).

$d_{j(i)}$, effect of jth dam mated to ith sire on its progeny, due primarily to genetic causes, but also to non-genetic maternal effects NID (O, O^2_d).

$f_{k(ij)}$, effect of environment and unpredictable genetic factors effecting kth offspring from jth dam and ith sire NID (O, O^2_f).

All terms except μ are uncorrelated random variables with (O, O^2).

$(X_{ijk}, X_{ijk'})$ are full sibs for $(k \neq k')$

$(X_{ijk}, X_{ij'k'})$ are half sibs for $(j \neq j')$

$(X_{ijk}, X_{i'j'k'})$ are uncorrelated for $(i \neq i')$

Using the above model, it is proposed to estimate the genetic parameters and their inter-relationships, and thus get the complete genetic structure of base population in static condition (Pushkar Nath Bhat, Win Moi Tait and S. S. Prabhu).